

One Private App for a Family's Money, Memory and Work

Product story · feature highlights · engineering decisions

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Prasanaa Kartick T — Personal App Project

Flutter / Dart · Android

Android

Flutter · Dart

Offline-first

Encrypted backups

1. At a Glance

DepoVaultTick is a personal Android app that keeps a household's financial records, daily expenses, notes, secrets and work paperwork in one place — entirely on the phone, behind its own locks — and warns the owner in red and yellow before anything matures, falls due or expires.

5

main sections

6

dashboards

3

lock secrets

77

source files

91k

lines of Dart

Project type	Personal app, designed, built and documented end to end by one developer
Platform	Android
Built with	Flutter and Dart
Data model	Offline-first — everything is stored on the device; no accounts, no cloud
Reports	Interactive dashboards with charts; Excel, CSV and PDF output
Protection	Three-secret App Lock, separate Stash Lock, optional biometrics, password-encrypted backups

2. The Problem

A typical household's important details are scattered: fixed-deposit receipts in a drawer, insurance premiums remembered by habit, Post Office passbooks in a bag, SIP dates in a bank app, certificates that quietly expire, and passwords in a notebook. Missing a single date can cost money or cover.

- Information is spread across paper, many apps and memory.
- Nothing tells you what is coming up **across** all of it.

- Typical tools want an account, a cloud upload and a subscription — a poor fit for sensitive family data.
- The date rules are fiddly: recurring instalments, paused plans, extended policy terms, month-end dates.

3. The Solution

One app with five sections, a single combined alert list, and one consistent set of colour-coded warnings.

Section	What it gives the user
Family & Vault	Per-person tracking of bank deposits (fixed and recurring), insurance policies, Post Office schemes and mutual funds, with maturity and due-date warnings and family-wide dashboards
Daily Log	An expense register, a bill and reminder board, and a calendar that also shows what is coming up in every other section
Notes	A corkboard of coloured sticky notes with real text formatting
Stash	A small private safe for passwords and PINs, behind its own lock
Work	A day-by-day invoice and PDF generator, a contact book, and a certificate tracker with expiry reminders

4. Feature Highlights

Never miss a date

- A bell with six periods — Today, Tomorrow, Next 7 Days, This Month, This Quarter and Missed — that combines every date-based item in the app.
- **Red means act now, yellow means due soon.** The most severe colour rolls up from one deposit to its bank, the person, the whole family and the Home logo.
- Missed instalments and premiums can be ticked off, with up to five steps of undo.

See the whole family at once

- A rotating “Next Up” panel shows the single most urgent item across every person and every category.
- Four financial dashboards with filters, charts, calendars and detailed tables, plus two more for daily expenses and invoices.
- Every dashboard exports to Excel or CSV through the phone’s Share menu.

Private by design

- No accounts and no servers: data never leaves the phone unless the owner exports it.

- An App Lock that asks for one of three different secrets at random, and a separate lock for the Stash.
- Fingerprint or face unlock as an optional faster route in.

Take your data with you

- A password-encrypted backup file with a choose-what-to-include checklist, down to individual family members.
- A guided restore with a **Merge** mode that shows any conflicting records side by side before anything is saved.

Built for daily life and work

- Rich-text sticky notes in six colours.
- A five-step invoice wizard that turns a timesheet into a PDF, with a client-folder archive and a billing dashboard.
- A certificate tracker with nested folders and reminder windows.

5. A Day with the App

1. **Open.** A short branded splash fades into Home, which asks for one of the three lock secrets, picked at random.
2. **Glance.** The Home logo and bell are yellow: something is due tomorrow. Tapping the bell shows an unpaid insurance premium.
3. **Check.** In Family & Vault, the “Next Up” panel shows the same premium; the person’s page shows which of their four categories carries the badge.
4. **Log.** In Daily Log the owner records the day’s grocery spend and marks last month’s rent as paid.
5. **Capture.** A quick sticky note is saved to the corkboard.
6. **Bill.** Under Work, a month’s timesheet becomes a PDF invoice.
7. **Protect.** Before changing phones, the owner exports an encrypted backup, to be restored later with Merge or Replace.

6. Engineering Highlights

The parts of the project that took the most thought, and the reasoning behind each.

Challenge	Approach	Why it matters
Totals that never go stale	Totals, statuses and due dates are recomputed from raw facts every time rather than stored	No chance of a saved number disagreeing with the records

Challenge	Approach	Why it matters
Recurring Deposits have no “paid so far” field	Count the instalment dates that have passed since the start date and multiply by the instalment amount	Correct figures from very little stored data
Paused SIPs	Stopped windows are skipped when counting, and the whole schedule shifts later by the pause length	A 12-instalment plan still completes all 12
Dashboards drifting from the real rules	Dashboards call the same shared status functions the main screens use	One source of truth; a policy can’t look different in two places
Bell versus list disagreeing	Home uses a lightweight check that mirrors the exact rules of the bell’s tabs	The Home badge always matches the list behind it
Shoulder-surfing a PIN	Three secrets set up together; one asked at random; keypad reshuffled each time; five-try lock-out	Seeing one unlock isn’t enough to get in next time
Merging a backup safely	Conflicting records are shown side by side; Save stays disabled until every conflict has a decision	Nothing is silently overwritten
Repeating bills	Each settled occurrence is recorded by its own date; unpaid ones appear as separate “missed” lines	Two unpaid months show as two rows, not one
Rich text in a plain text box	Text is stored as short styled “runs” that join back into the exact sentence	Bold, italic and underline without a heavy editor
Hand-correctable certificate expiry	Expiry is a stored field, pre-filled from acquired date + validity, and never silently recalculated once overridden	A manual correction always sticks
Overnight shifts	If the end time isn’t later than the start, it is rolled into the next day	Hours never come out negative
Excel needs a dropdown the library can’t make	The finished file is adjusted internally, with safety checks that fall back to the plain file	A richer export that can’t be corrupted
Month-end dates	A date that doesn’t exist in the target month is clamped to its last day	The 31st behaves sensibly in February

7. Design Decisions

- **Offline-first.** Privacy and reliability came before convenience features such as sync.
- **Warn early, warn consistently.** One red / yellow language is used on tiles, rows, folders, the bell and the logo.

- **Optional fields.** Insurance and Post Office entries need only a policy or account number, because real paperwork is filled in gradually.
- **Double confirmation for destructive actions.** Deleting a person or a bank, which also deletes everything under it, asks twice.
- **One look and feel.** A shared header, footer, date picker, clock picker, suggestion box and sound feedback are reused on every screen.
- **Sound with meaning.** A plain click for navigation and a distinct chime when something is saved, with one switch to mute the whole app.

8. Security and Privacy Approach

Layer	What it does
App Lock	Guards the whole app on every open and on return from the background
Stash Lock	A separate PIN-and-pattern lock, asked afresh at each sensitive moment
Biometrics	Optional; the app only receives a yes or no from the phone and never sees biometric data
Stored lock codes	Kept only in scrambled form, never in plain text
Backups	Encrypted with a password the owner chooses; a forgotten password cannot be recovered, by design
No network dependency	The app has no accounts or servers, so there is no remote copy to leak

An honest limit

The locks protect against someone holding the phone. They complement the phone's own screen lock rather than replace it, and an exported file leaves the app's protection behind, which is why backups carry their own password.

9. Technology

Area	Choice
Framework	Flutter with Dart, Material 3 styling
Platform	Android
Storage	On-device saved settings behind one central storage service
Charts	A charting library for every dashboard
Documents	PDF generation for invoices and receipts; Excel and CSV generation for exports
Security	Encryption and hashing libraries; the phone's own biometric system

Area	Choice
Media	A sound player for interface feedback; bundled logo, artwork and font assets
Sharing	The phone's native Share menu and file picker

10. What This Project Demonstrates

- **Product thinking** — a clear problem, a defined user, and a single consistent experience.
- **Data modelling** — nested and flat records, linked by identifiers, with all totals derived from the facts.
- **Date and time logic** — recurring schedules, pauses, month-end handling and overnight shifts.
- **User experience** — warnings, rolling summaries, inline pickers and forgiving forms.
- **Security thinking** — layered locks, safe failure modes and honest limits.
- **Data portability** — encrypted backup, merge-with-review restore, and Excel, CSV and PDF output.
- **Scale and consistency** — about 91,000 lines across 77 files that still behave like one app.
- **Documentation** — 17 plain-English guides that explain each part without exposing the code.

11. Where It Could Go Next

Ideas for future versions, not commitments:

- A store-ready release configuration and final app listing.
- An iPhone version, since the app is built on a cross-platform framework.
- Optional encrypted cloud sync for owners who want it.
- Native phone notifications for upcoming dates, alongside the in-app bell.
- Additional Post Office schemes, insurance types and currencies.

12. Companion Guides

Each part of the app has its own plain-English guide:

Guide	Covers
App Overview	The whole app on one page
Setup & Startup · Home Screen	Foundations and the hub
Notifications & Alerts · App Lock & Privacy	The warning system and the locks
Family & Banks · Deposits · Insurance · Post Office · Mutual Funds	The financial tracking section, one guide each

Guide	Covers
Daily Log · Notes · Stash	The everyday sections
Work — Invoices · Work — Rolodex & Certificates	The work tools
Backup, Export & Import · Shared Building Blocks	Data portability and common parts